

Mode 5 — Baseline-BFP

DP32 (8×4) → DP16 (8×4) → DP8 (8×4) → block exponents

Step 1 — DP32 (8×4)

$$\text{DP32 } (8 \times 4) = \sum_{i=0}^{31} a_i b_i$$

Step 2 — DP16 (8×4)

$$\text{DP32 } (8 \times 4) = \underbrace{\sum_{i=0}^{15} a_i b_i}_{\text{DP16 } (8 \times 4)} + \underbrace{\sum_{i=16}^{31} a_i b_i}_{\text{DP16 } (8 \times 4)}$$

Step 3 — DP8 (8×4)

$$\text{DP32 } (8 \times 4) = \underbrace{\sum_{i=0}^7 a_i b_i}_{\text{DP8 } (8 \times 4)} + \underbrace{\sum_{i=8}^{15} a_i b_i}_{\text{DP8 } (8 \times 4)} + \underbrace{\sum_{i=16}^{23} a_i b_i}_{\text{DP8 } (8 \times 4)} + \underbrace{\sum_{i=24}^{31} a_i b_i}_{\text{DP8 } (8 \times 4)}$$

Step 4 — Block exponents (one exponent block per DP8 (8×4))

$$\left(\sum_{i=8q}^{8q+7} a_i b_i, E_q \right)_{q=0..3} \longrightarrow \left(M, E \right) \quad E_q = E_{a,q} + E_{b, \lfloor q/2 \rfloor}$$

$E = \max(E_0, \dots, E_3)$, $\delta_q = E - E_q$ (staged in the tree, the stage shifts telescope); truncating shifts, on block sums only;
 $\text{DP32 } (8 \times 4) = M \cdot 2^E$.

Step 5 — Result

$$M = 2^{-\delta_0} \sum_{i=0}^7 a_i b_i + 2^{-\delta_1} \sum_{i=8}^{15} a_i b_i + 2^{-\delta_2} \sum_{i=16}^{23} a_i b_i + 2^{-\delta_3} \sum_{i=24}^{31} a_i b_i, \quad E = \max(E_0, E_1, E_2, E_3)$$